



Cyber Security

Project 1

Industrial Training Kit

Batch: 2026 | Powered by DecodeLabs

WELCOME TO THE TEAM! 🚀

Step into the role of a Cybersecurity Analyst at DecodeLabs. Project 1 is your defensive phase: The Password Strength Checker. This track isn't about "complex hacking"—it's about Security Logic. Before you protect massive networks, you must master the fundamental principles of data validation and entropy. By completing this milestone, you are proving you can build a program that evaluates risk through pure string-handling and conditional logic. Let's design a digital foundation that safeguards user identity with absolute precision.



GUIDELINES:

Qualification Criteria

EARN YOUR BADGE 

- Requirement: Complete project 1 to unlock other projects for next week.
- Standard: All projects must be verified for quality.
- Mandatory: Project 1 is the essential start for every intern.



Project 1:

Password Strength Checker



Goal:

Create a program that checks whether a password is weak, medium, or strong.

Key Requirements:

- Check password length
- Check use of numbers, symbols, and uppercase letters
- Display password strength result

Key Skills:

String handling, condition checks, security basics

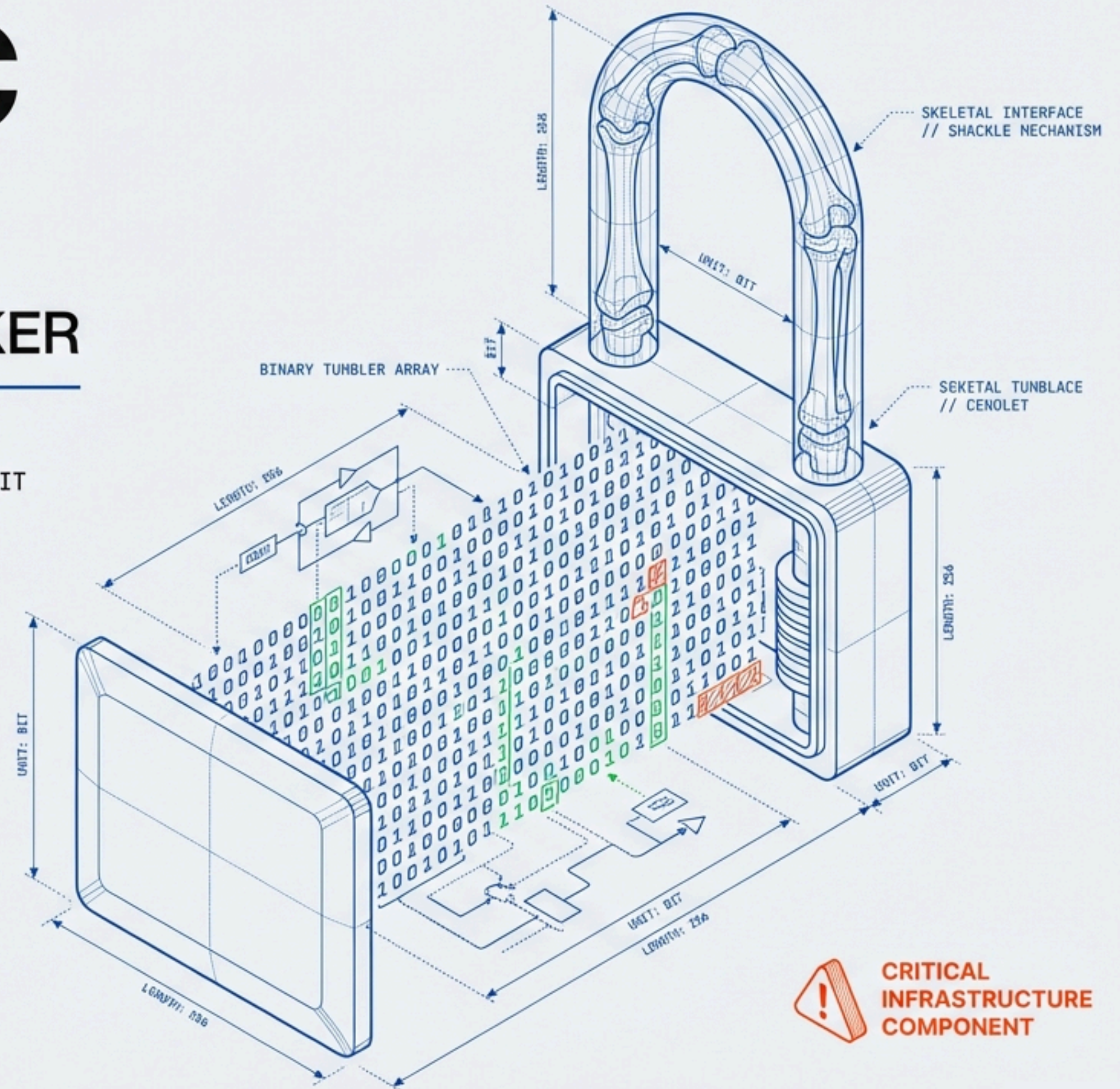
THE LOGIC SKELETON

PASSWORD STRENGTH CHECKER

PROJECT 1:
DEFENSIVE LOGIC

STATUS:
INITIALIZATION

TRACK:
JUNIOR ANALYST //
DECODELABS INDUSTRIAL KIT



THE STRATEGIC LANDSCAPE

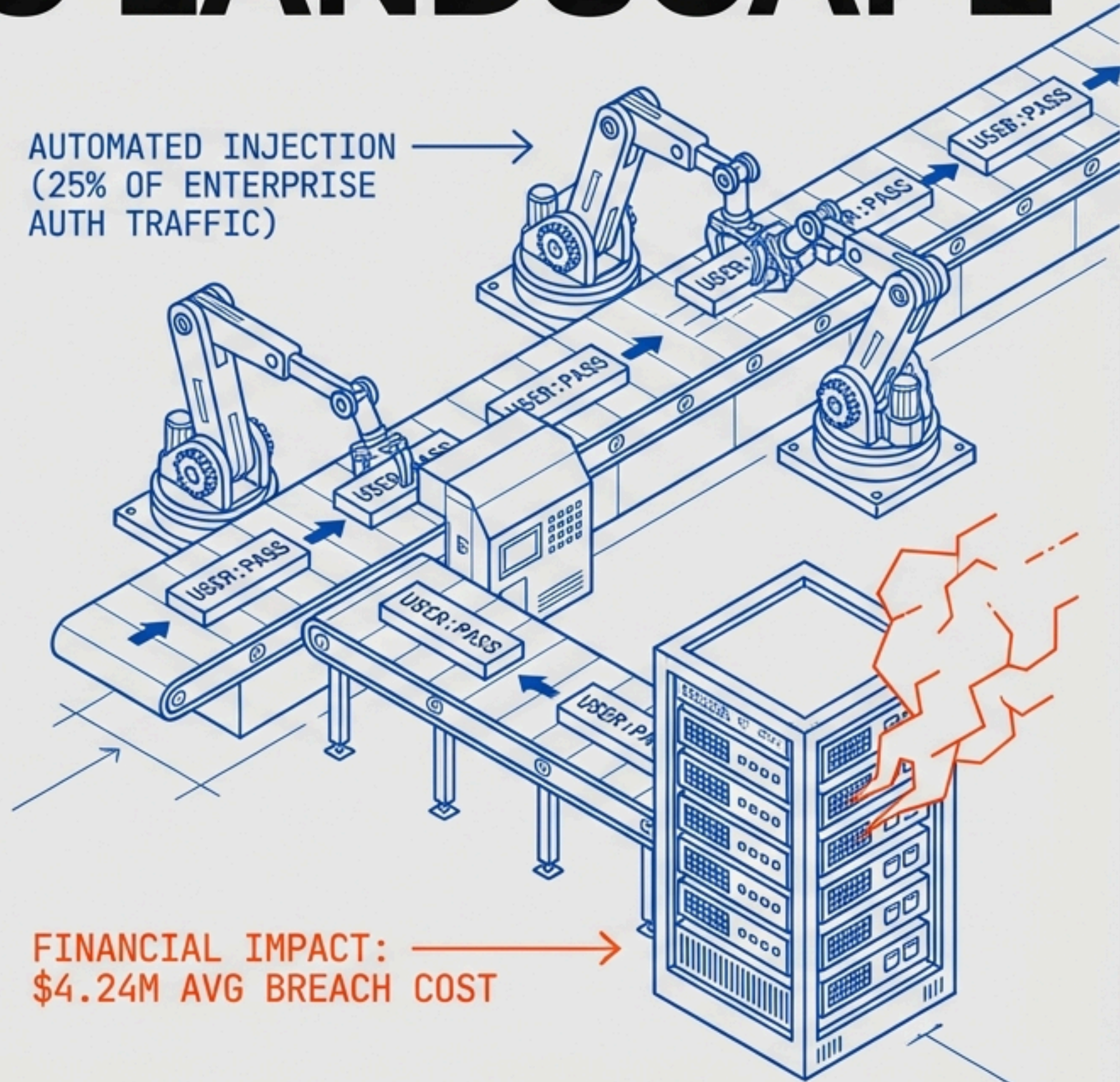
81%

JetBrains Mono Regular in Ink Black

OF HACKING-RELATED BREACHES
LEVERAGE WEAK OR STOLEN PASSWORDS.

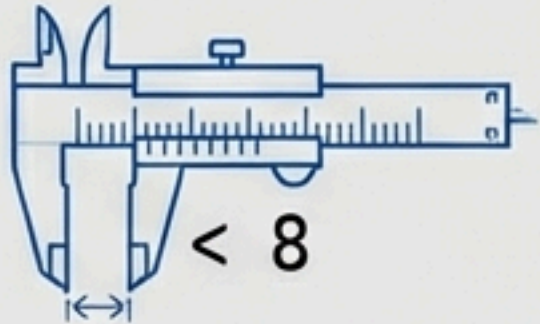
Source: Verizon DBIR

AUTOMATED INJECTION
(25% OF ENTERPRISE
AUTH TRAFFIC)



FINANCIAL IMPACT:
\$4.24M AVG BREACH COST

DEFINING THE POLICY: THE ZERO POINT



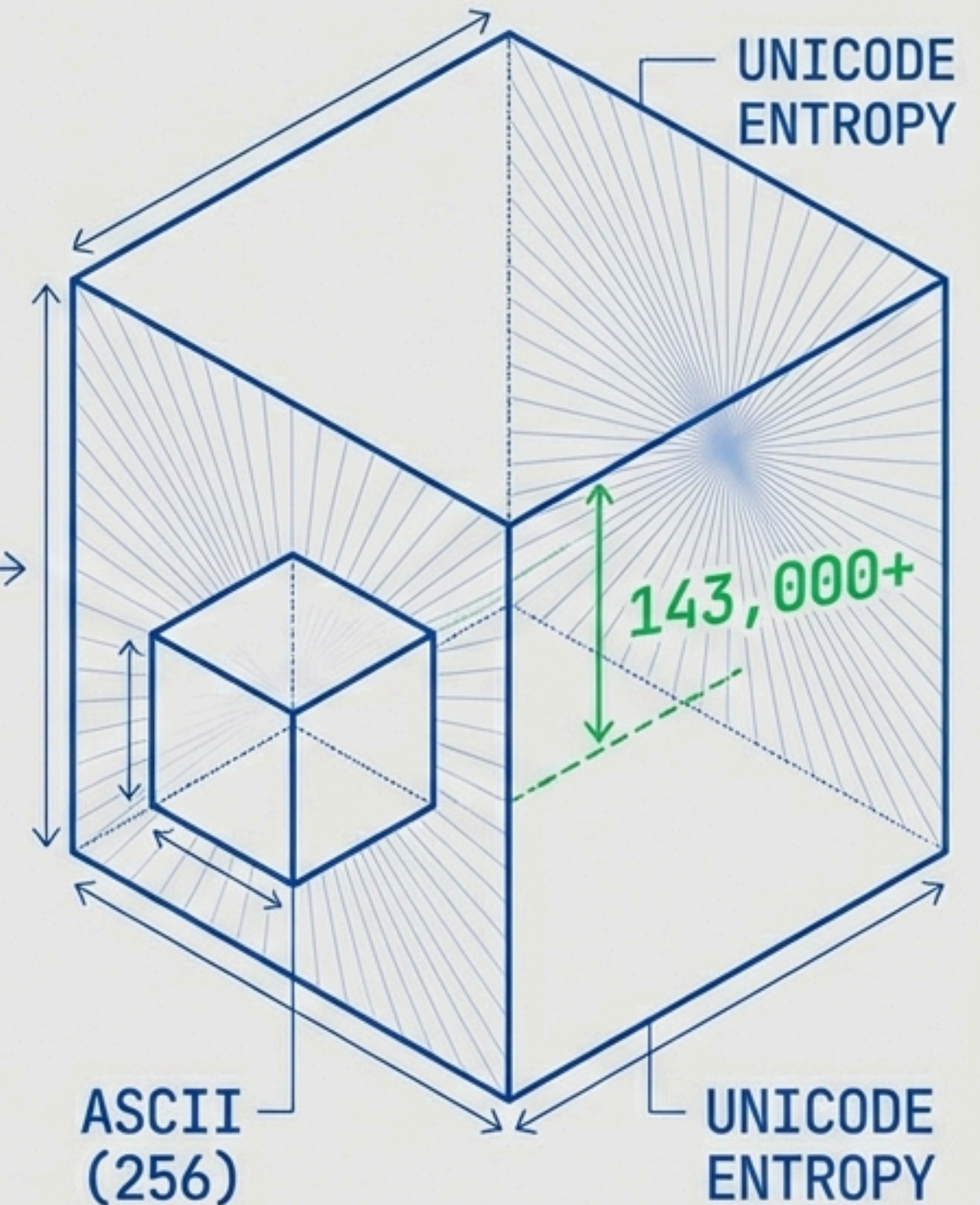
LENGTH VERIFICATION:
< 8 CHARS = IMMEDIATE FAIL.
(EXPONENTIAL BRUTE FORCE RISK)

A	9	#	*	!
A	9	#	@	*
A	9	#	@	*
A	9	@	*	!

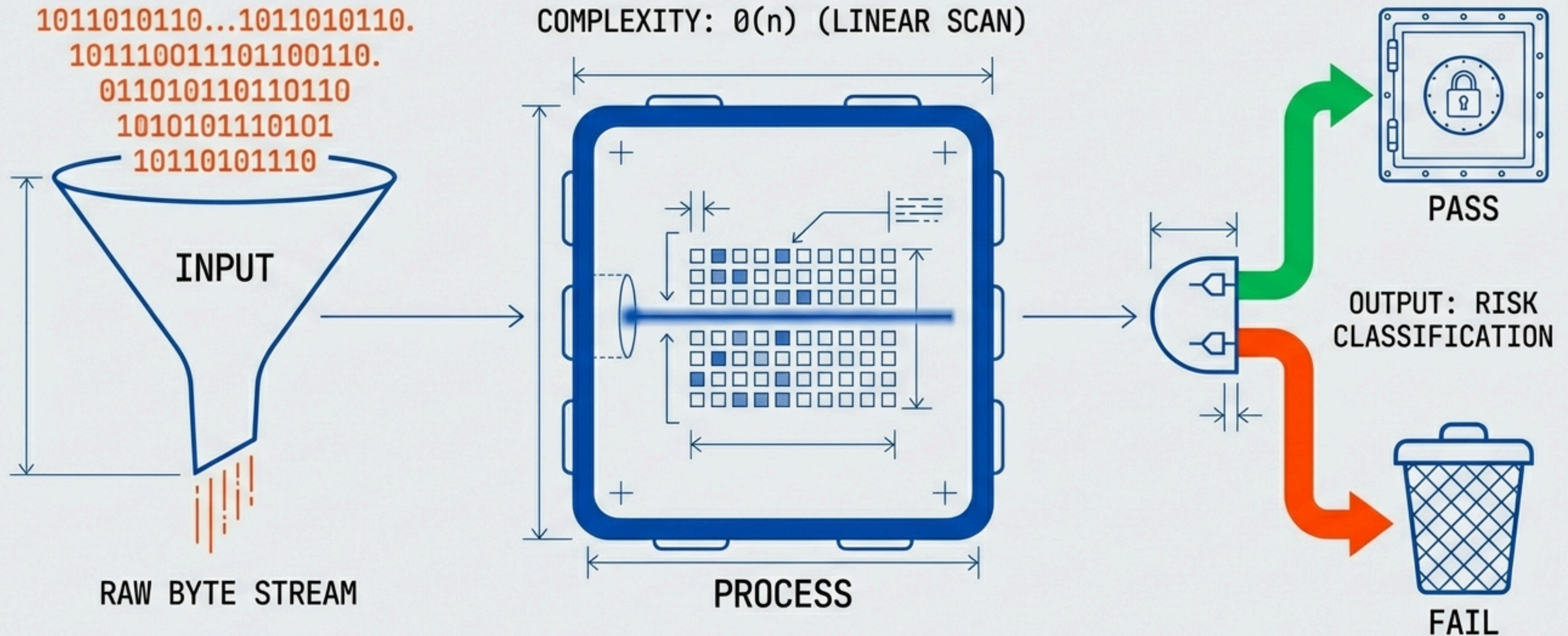
PATTERN RECOGNITION:
MANDATORY [A-Z], [0-9],
[SYMBOLS].



THE UNICODE CURVEBALL:
EXPAND SEARCH SPACE FROM 95
(ASCII) TO 143,000+ (UNICODE).



THE LOGIC SKELETON: IPO MODEL

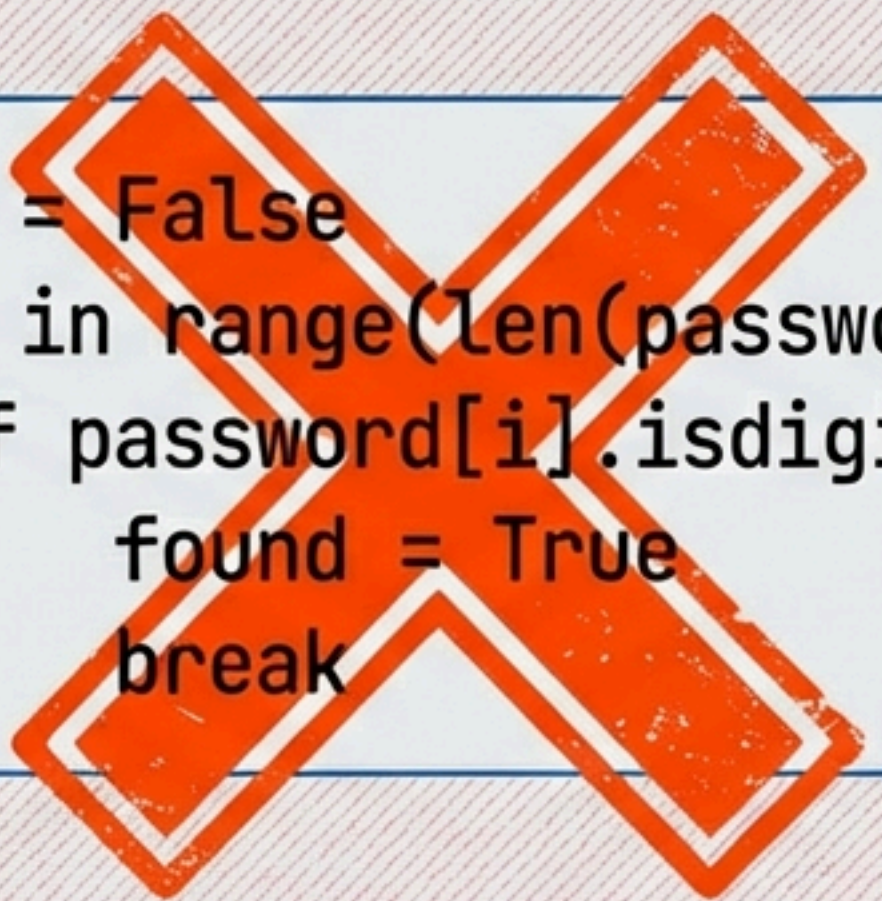


CONSTRAINT: VALIDATION TIME MUST GROW LINEARLY, NOT EXPONENTIALLY.

COMPUTATIONAL EFFICIENCY: PYTHONIC ELEGANCE

THE AMATEUR: VERBOSE APPROACH

```
found = False
for i in range(len(password)):
    if password[i].isdigit():
        found = True
        break
```



MANUAL LOOP (VERBOSE, SLOW)

THE PROFESSIONAL: PYTHONIC APPROACH

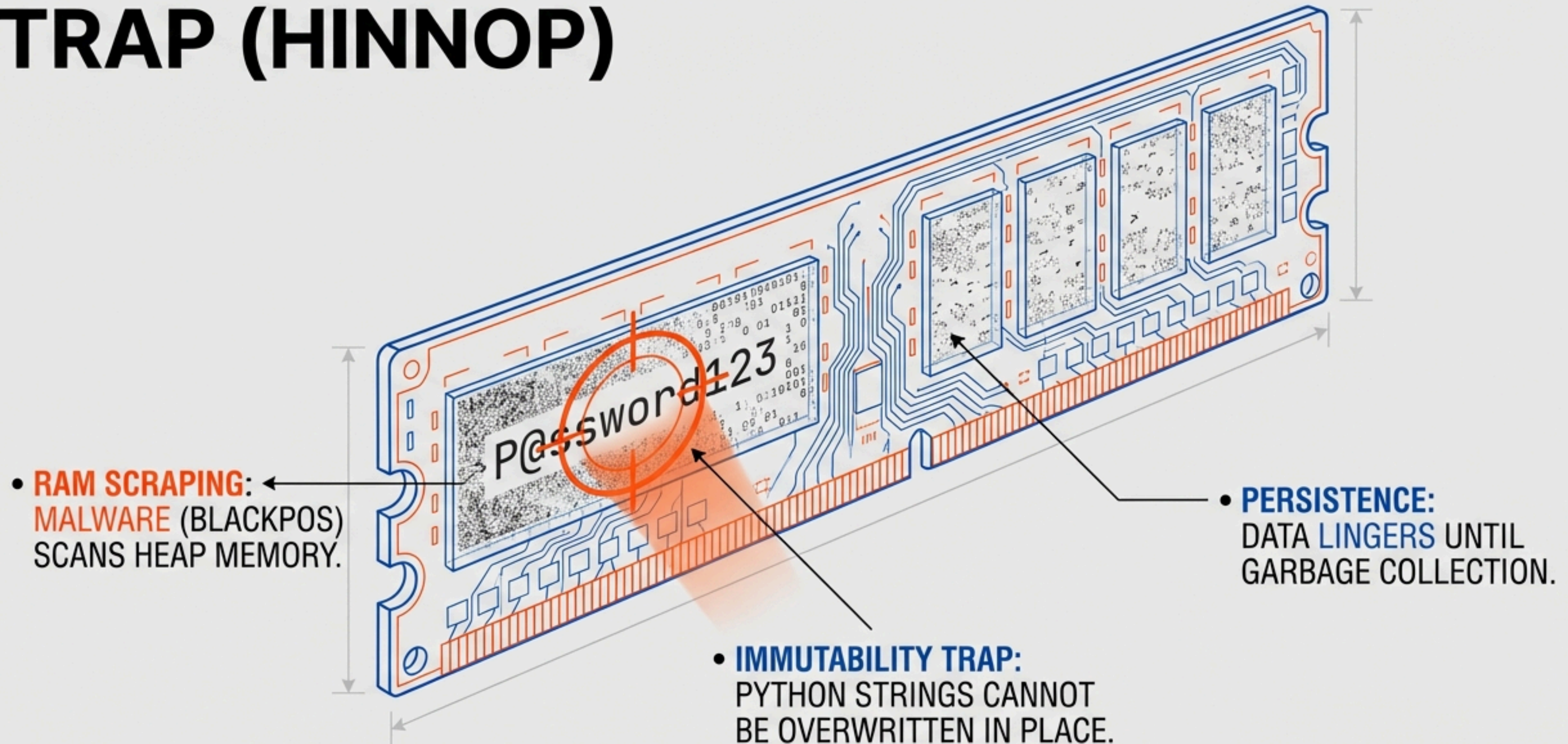
```
has_digit = any(char.isdigit()
                 for char in password)
```



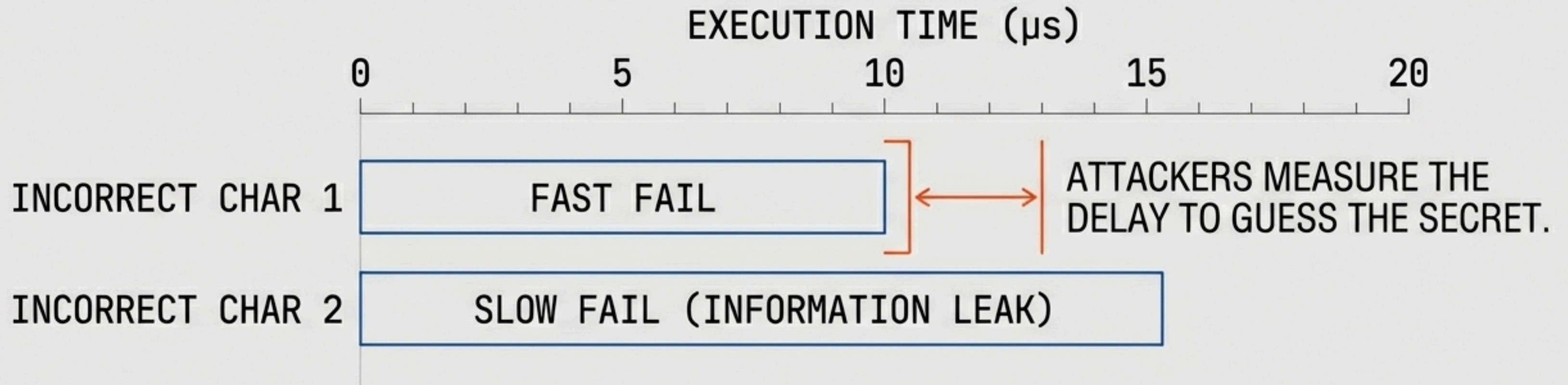
1. C-OPTIMIZED
BUILT-IN

2. SHORT-CIRCUIT
EXECUTION

THE VOLATILE SECURITY TRAP: DATA IN RAM TRAP (HINNOP)



ADVANCED VULNERABILITY: TIMING ATTACKS

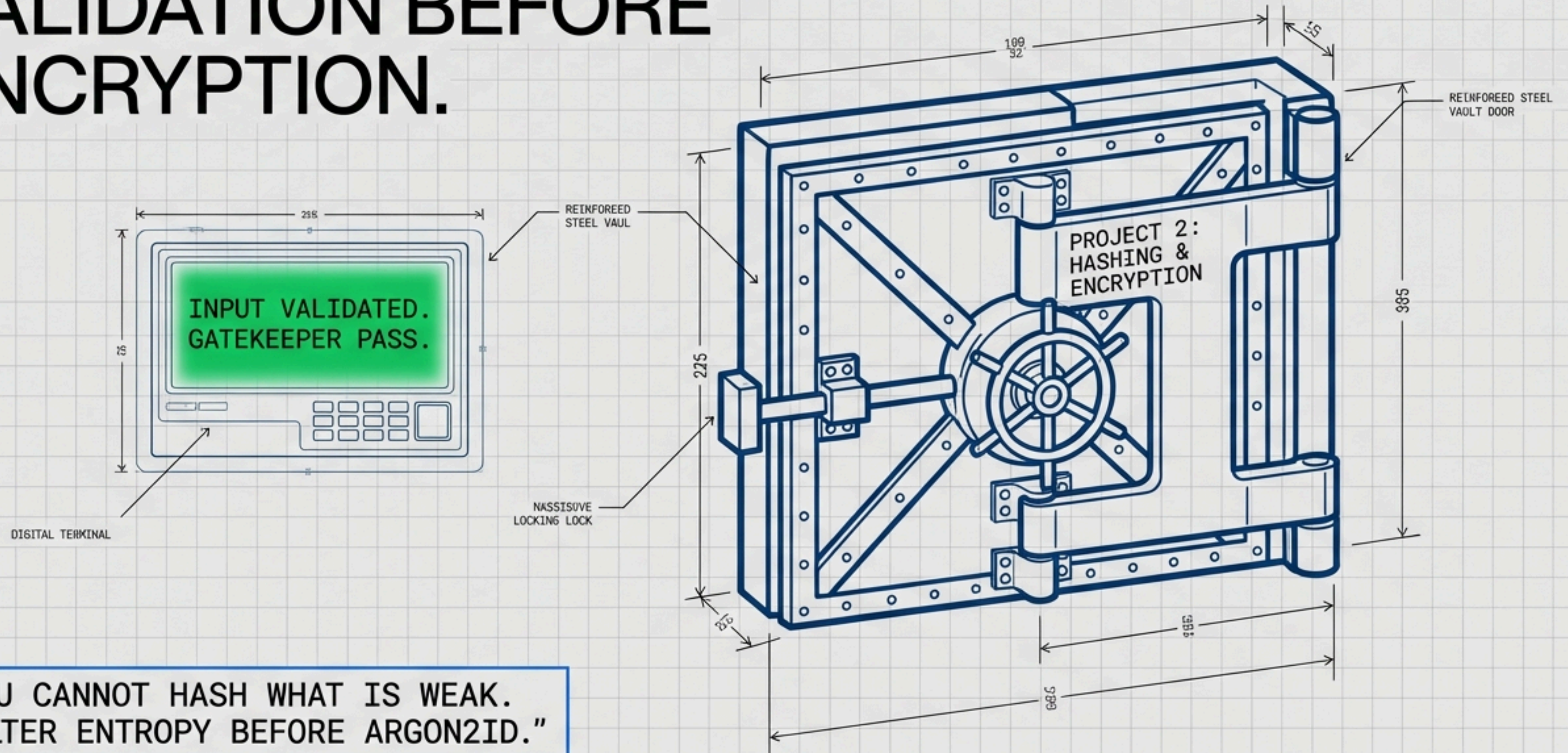


SOLUTION:

USE: `hmac.compare_digest(a, b)` ←
↑
CONSTANT TIME COMPARISON (OPENSSL CRYPTO_MEMCMP)

THE GATEKEEPER RULE

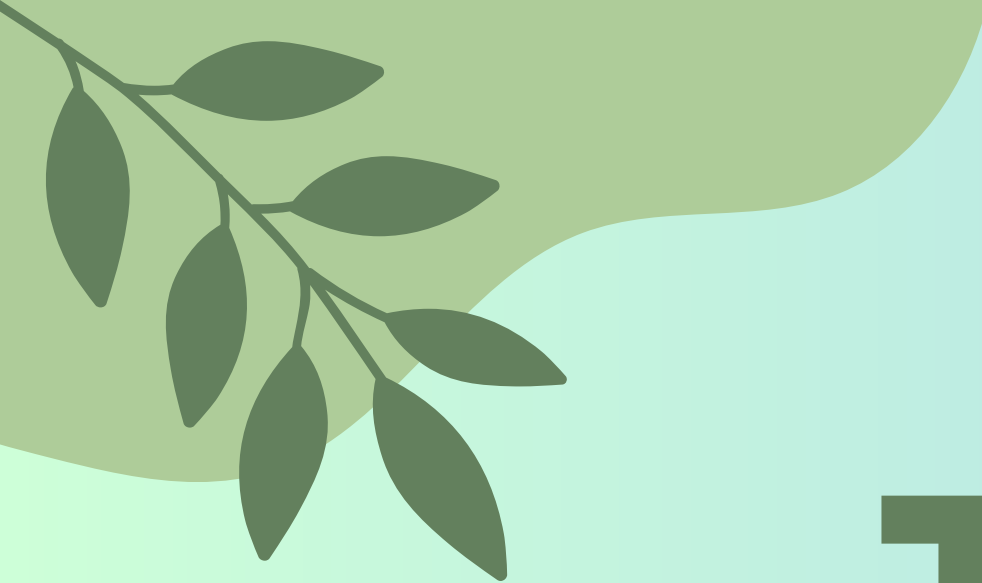
VALIDATION BEFORE ENCRYPTION.



"YOU CANNOT HASH WHAT IS WEAK.
FILTER ENTROPY BEFORE ARGON2ID."

CONCLUSION

The absolute best way to master Cyber security is through hands-on practice, not just theory. Don't just aim to complete these projects; take them one by one, experiment with unique solutions—like adding a check for common "leaked" passwords or implementing character variety requirements—and treat every logic error as a learning opportunity. As you build these milestones at DecodeLabs, you are creating a real-world portfolio that showcases your security mindset to future employers. Your journey to becoming a professional security expert begins right here, right now, with the very first line of code you write today.



THANK YOU

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 GREATER LUCKNOW, INDIA

